



PY004(/M), PY004Z9(/M) and PY004VX(/M) High-Load Pitch and Yaw Platforms

User Guide

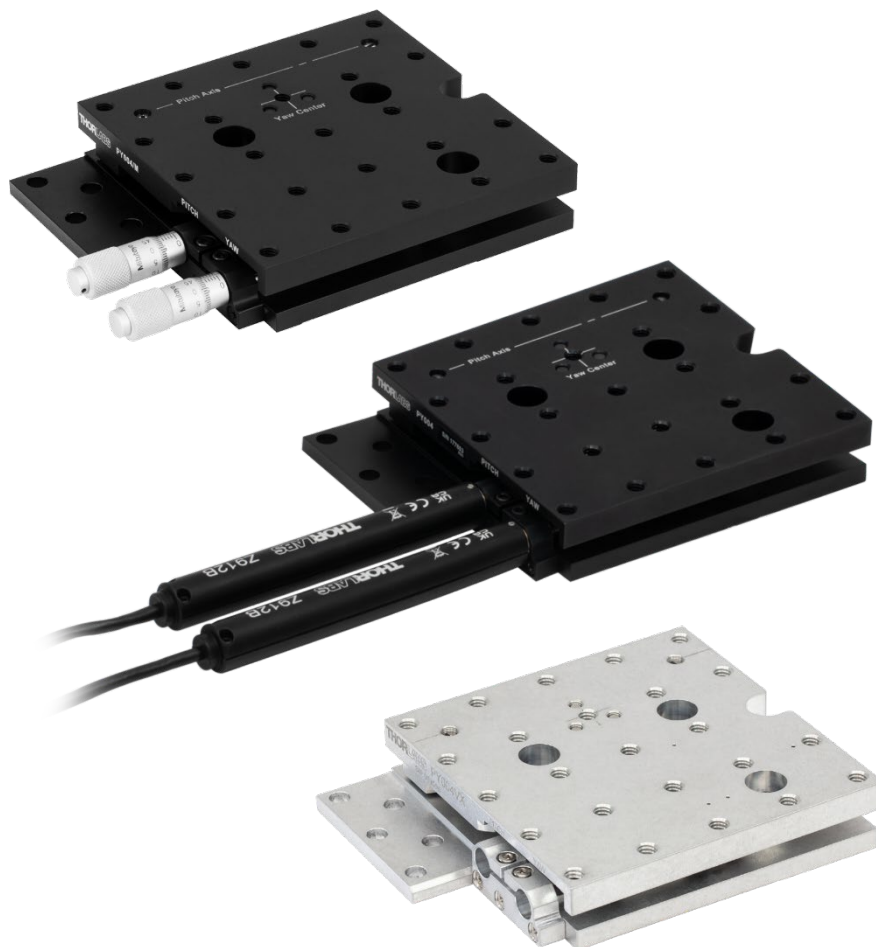


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Chapter 1 Introduction

1.1 Intended Use

The PY004 High Load Pitch and Yaw Platforms are high precision devices to precisely control the rotational movement of an object around two axes: pitch and yaw.

They are intended for indoor use only and can be used in, for example, optical, robotics and instrumentation environments. They should not be used in explosive environment. To ensure reliable operation the unit should not be exposed to corrosive agents or excessive moisture, heat or dust. The vacuum version can be used in vacuum environments.

NOTICE

Impairment of protection and warranty

If this equipment is used in a manner not specified by the manufacturer, its protection may be impaired. The product may only be used in accordance with the instructions described in this manual. Any other use will invalidate the warranty.

1.2 Explanation of Safety Warnings

NOTICE

Indicates information considered important, but not hazard-related, such as possible damage to the product.



The CE/UKCA markings on the product are the manufacturer's declaration that the product complies with the essential requirements of the relevant European health, safety, and environmental protection legislation.



The symbol on the product, the accessories or packaging indicates that this device must not be treated as unsorted municipal waste but must be collected separately.

1.3 Description

The PY004 Pitch and Yaw Platforms provide $\pm 2.5^\circ$ of adjustment in pitch and $\pm 4.00^\circ$ in yaw. They are designed for use with up to 5 kg (11 lbs) loads, such as lasers, cameras and 3-Axis stages (MBT6xx, MAX3xx). Their top platform features an array of 24 1/4"-20 (M6 x 1.0) threaded mounting holes on a 1" (25 mm) pitch. The PY004(/M) stage is fitted with high-resolution micrometers that are marked to provide positioning accuracy within 9 arcseconds in pitch and 16.2 arcseconds in yaw. Turning the micrometers gives 0.25° of pitch and 0.45° of yaw per revolution.

Thorlabs offer the following options:

- PY004(/M)
Pre-configured manually operated device with high-resolution micrometers.
- PY004Z9(/M)
Pre-configured motorized device utilizing two pre-fitted Z912B DC servo actuators.
- PY004VX(/M)
User-configurable vacuum compatible version (no adjusters included).

NOTICE

Adjusters are not provided with the PY004VX(/M) but can be ordered separately. Compatible platform/actuator combinations are shown in Section 1.4.1.

The use case application in Figure 1 shows a pair of MAX312D Stages, each mounted to a PY004 stage. This arrangement gives pitch and yaw adjustment of the fiber to aid in optimizing the fiber coupling.

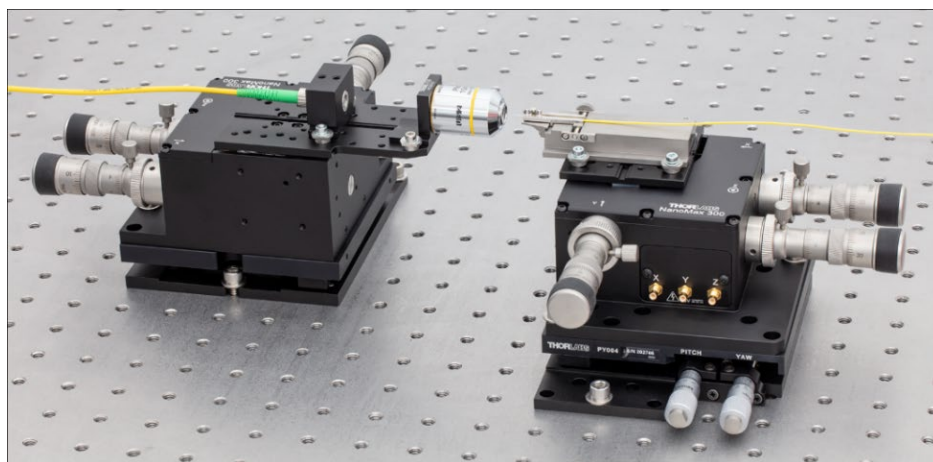


Figure 1 PY004 Typical Application

Figure 2 shows the PY004Z9(/M) stage fitted with a pair of Z912B DC motor actuators. This arrangement can be driven by a pair of KDC101 motor controllers.



Figure 2 PY004Z9 Motorized Stage

1.4 Technical Data

1.4.1 Specifications

Item #		PY004(/M)
Adjustment Range		Pitch $\pm 2.5^\circ$ Yaw: $\pm 4.0^\circ$
Minimum Incremental Movement ^a		Pitch: 9 arcsec Yaw: 16.2 arcsec
Crosstalk		$<0.05^\circ$ (3 arcmin)
Weight		679 g (1.5 lbs)
Horizontal Load Capacity (Max) ^b		5.0 kg (11.0 lbs)
Vertical Load Capacity (Max), by Distance from Top Platform ^c	30 mm (1.18")	1.8 kg (4.0 lbs)
	50 mm (1.97")	1.1 kg (2.4 lbs)
	80 mm (3.15")	0.7 kg (1.5 lbs)
Deck Height		25.0 mm (0.98")
Bushing Diameter		9.5 mm (3/8")
Construction		Black Anodized Aluminum
Operating Temperature		5° C to 40° C (41° F to 104° F)
Included Drives		High resolution micrometers (Qty. 2)

- The minimum incremental movement is based on half a graduation of the micrometer scale.
- The load capacity when the stage's base is mounted on a horizontal surface. The max load capacity is different at certain points of the platform. Refer to spatial drawings Figure 4 and Figure 5 for specific load capacity information.
- The load capacity when the stage's base is mounted on a vertical surface. The load capacity will depend on the stage's orientation, the positioning of the load on the top platform, and the distance of the load from the platform. The load data provided is approximate and meant as a guide only.

Item #		PY004Z9(/M)
Adjustment Range		Pitch $\pm 2.5^\circ$ Yaw: $\pm 4.0^\circ$
Minimum Incremental Movement		Pitch: 7.13 arcsec Yaw: 15.71 arcsec
Bi-directional Repeatability		Pitch: 27.85 arcsec Yaw: 4.75 arcsec
Maximum Velocity		Pitch: 3600 arcsec/s Yaw: 3600 arcsec/s
Maximum Acceleration		Pitch: 3600 arcsec/s ² Yaw: 3600 arcsec/s ²
Crosstalk		<0.05° (3 arcmin)
Weight		870 g (1.91 lbs)
Horizontal Load Capacity (Max)^{a,b}		2.0 kg (4.4 lbs)
Vertical Load Capacity (Max), by Distance from Top Platform^c	30 mm (1.18")	1.8 kg (4.0 lbs)
	50 mm (1.97")	1.1 kg (2.4 lbs)
	80 mm (3.15")	0.7 kg (1.5 lbs)
Deck Height		25.0 mm (0.98")
Bushing Diameter		9.5 mm (3/8")
Construction		Black Anodized Aluminum
Operating Temperature		5° to 40° C (41° to 104° F)
Included Drives		Z912B DC Servo Actuators (Qty. 2)

- The load capacity when the stage's base is mounted on a horizontal surface. The max load capacity is different at certain points of the platform. Refer to spatial drawings Figure 4 and Figure 5 for specific load capacity information.
- The Z912B(V) actuator could wear out after approximately 5000 cycles if used continuously at the max load
- The load capacity when the stage's base is mounted on a vertical surface. The load capacity will depend on the stage's orientation, the positioning of the load on the top platform, and the distance of the load from the platform. The load data provided is approximate and meant as a guide only.

Item #		PY004VX(/M) – No Actuators Included	
Adjustment Range		Pitch $\pm 2.5^\circ$ Yaw: $\pm 4.0^\circ$	
Crosstalk		<0.05° (3 arcmin)	
Weight		622 g (1.37 lbs)	
Vertical Load Capacity (Max), by Distance from Top Platform ^a	30 mm (1.18")	1.8 kg (4.0 lbs)	
	50 mm (1.97")	1.1 kg (2.4 lbs)	
	80 mm (3.15")	0.7 kg (1.5 lbs)	
Vacuum Rating ^b		10 ⁻⁶ Torr	
Deck Height		25.0 mm (0.98")	
Bushing Diameter		9.5 mm (3/8")	
Construction		Unanodized Aluminum	
Maximum Bakeout Temperature		With vacuum adjusters^c: 130 °C (266 °F) With PIA13VF actuators: 130 °C (266 °F) With Z912BV actuators: 85 °C (185 °F)	
Operating Temperature		5° C to 40° C (41° F to 104° F)	
Included Drives ^d		N/A	

- a. The load capacity when the stage's base is mounted on a vertical surface. The load capacity will depend on the stage's orientation, the positioning of the load on the top platform, and the distance of the load from the platform. The load data provided is approximate and meant as a guide only.
- b. Following appropriate processing and bakeout procedures.
- c. Made up of components F25USA1, LN25100, POLARIS-N5 and F25US200V.
- d. Drives are not supplied with the PY004VX(/M). It is recommended to add compatible drives to an order when making a purchase.

Compatible Actuators						
Item#	Actuator Type	Values with PY004VX(/M)				
		Minimal Incremental Movement	Bi-directional Repeatability	Maximum Velocity (arcsec/s)	Acceleration (arcsec/s ²)	Horizontal Load Capacity (Max) ^a
Z912B(V)	DC Servo	Pitch: 7.13 arcsec Yaw: 15.71 arcsec	Pitch: 27.85 arcsec Yaw: 4.75 arcsec	Pitch: 3600 Yaw: 3600	Pitch: 3600 Yaw: 3600	2.0 kg (4.4 lbs) ^b
PIA13(VF) ^c	Piezo Inertia	Pitch: 54 milliarcsec Yaw: 97.2 milliarcsec	-	27.7 - 83.1	-	5.0 kg (11.0 lbs)
Vacuum Adjuster ^d	Manual	Pitch: 457 arcsec/rotation Yaw: 823 arcsec/rotation	N/A	N/A	N/A	5.0 kg (11.0 lbs)
ZFS13B ^c	Compact Stepper	Pitch: 1 milliarcsec Yaw: 1.6 milliarcsec	-	2770	9000	1.8 kg (4.0 lbs)
ZST213B ^c	Stepper					

- a. The load capacity when the stage's base is mounted on a horizontal surface. The max load capacity is different at certain points of the platform. Refer to spatial drawings Figure 4 and Figure 5 for specific load capacity information.
- b. The Z912B(V) actuator could wear out after approximately 5000 cycles if used continuously at the max load.
- c. Calculated values.
- d. Made up of components F25USA1, LN25100, POLARIS-N5 and F25US200V.

NOTICE

Refer to the web presentations for complete specifications of actuators.

1.4.2 Mechanical Drawings

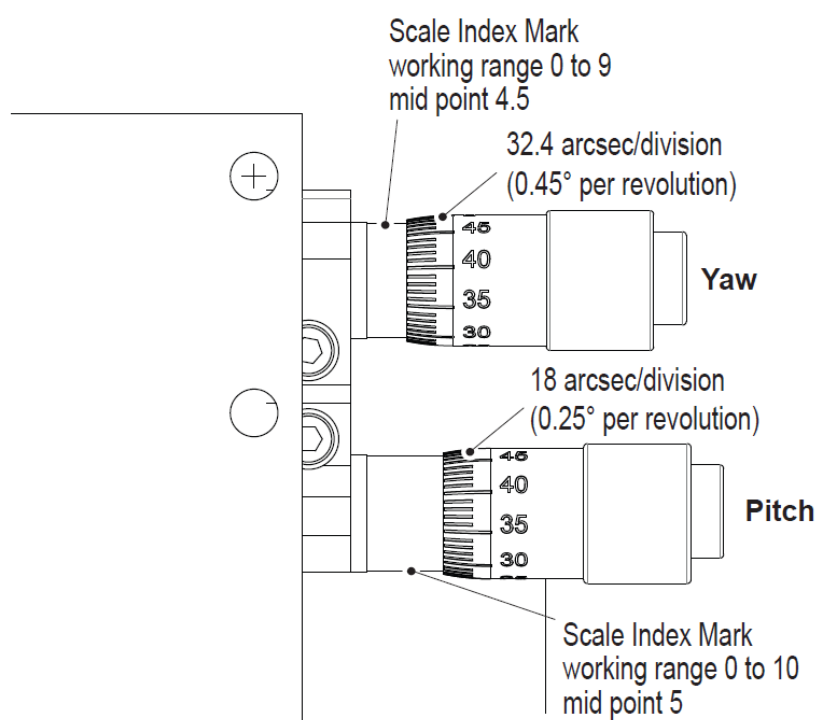


Figure 3 Reading Micrometer Drives

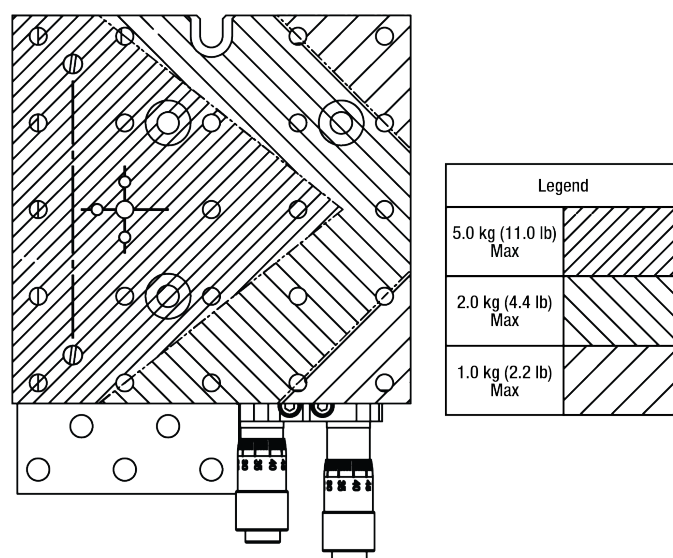


Figure 4 Spatial Load Specifications (Manual Stages)

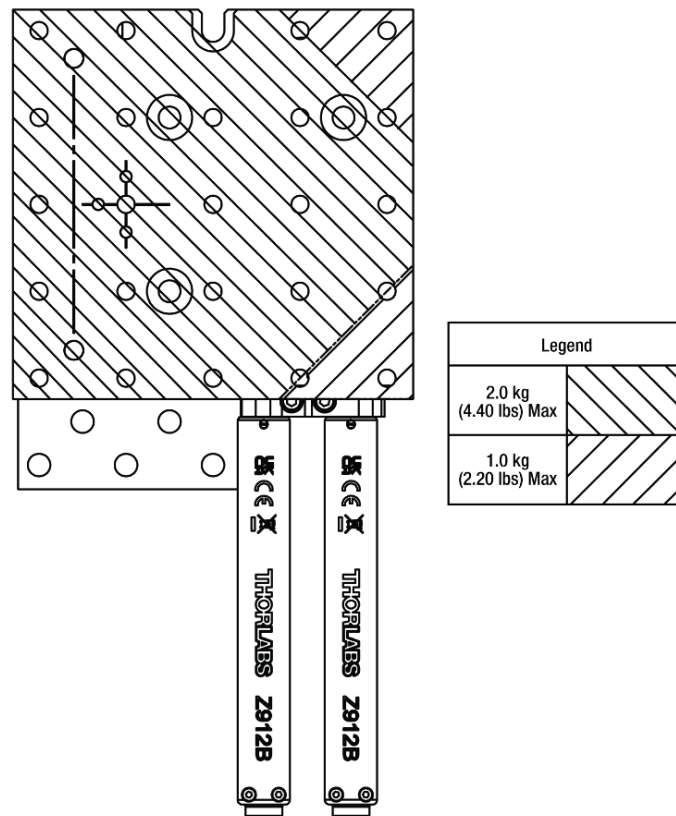


Figure 5 **Spatial Load Specifications (with Z912B actuators)**

1.5 Simplified Declaration of Conformity



EU Declaration of Conformity

Manufacturer: Thorlabs LTD

Address: 204 Lancaster Way, Ely, CB6 3NX

We hereby declare under our sole responsibility that:

Product: PY 004Z 9, PY 004Z 9/M

Product description: Motorised Pitch & Yaw Platform (Imperial and Metric)

is/are in conformity with the following directive(s):

2014/30/EU Electromagnetic Compatibility (EMC) Directive

2011/65/EU Restriction of Use of Certain Hazardous Substances (RoHS), including 2015/863

2006/42/EC Machinery Directive (MD)

and (harmonized) standards / technical specifications :

EN 61326-1 Electrical Equipment for Measurement, Control and Laboratory Use - EMC Requirements 2013

EN IEC 63000 Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances 2018

EN ISO 12100 Safety of Machinery. General Principles for Design. Risk Assessment and Risk Reduction 2010

I hereby declare that the equipment named has been designed to comply with the relevant sections of the above referenced specifications and complies with all applicable Essential Requirements of the Directives.

Signed:

Name: Keith Dhese

On: 31/10/2023

Position: General Manager





EU Declaration of Conformity

Manufacturer: Thorlabs LTD
Address: 204 Lancaster Way, Ely, CB6 3NX

We hereby declare under our sole responsibility that:

Product: PY004VX, PY004VX/M
Product description: Vacuum-Compatible Pitch and Yaw Platform

is/are in conformity with the following directive(s):

2011/65/EU Restriction of Use of Certain Hazardous Substances (RoHS), including 2015/863
2006/42/EC Machinery Directive (MD)

and (harmonized) standards / technical specifications :

EN IEC 63000	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances	2018
EN ISO 12100	Safety of Machinery. General Principles for Design. Risk Assessment and Risk Reduction	2010

I hereby declare that the equipment named has been designed to comply with the relevant sections of the above referenced specifications and complies with all applicable Essential Requirements of the Directives.

Signed:

Name: Keith Dhese

On: 5/8/2025

Position: General Manager



Chapter 2 Safety

NOTICE

Impairment of operation and damage to equipment

Excessive moisture may impair operation. Spillage of fluid, such as sample solutions, should be avoided. If spillage does occur, clean up immediately using absorbent tissue. Do not allow spilled fluid to enter the internal mechanism.

Damage to equipment

Spillage of fluid, such as sample solutions, should be avoided. If spillage does occur, clean up immediately using absorbent tissue. Do not allow spilled fluid to enter the internal mechanism.

Chapter 3 Installation

NOTICE

This precision device is only serviceable if returned and properly packed into the complete original packaging including the complete shipment plus the cardboard insert that holds the enclosed devices. If necessary, ask for replacement packaging. Refer servicing to qualified personnel.

3.1 Unpacking

1. Open the packaging carefully. Vacuum products should be unpacked in an appropriate environment with vacuum handling.

NOTICE

Retain the original packaging. The device is only serviceable if returned and properly packed into the complete original packaging including the complete shipment plus the cardboard insert that holds the enclosed devices.

NOTICE

Once removed from its packaging, the pitch and yaw platforms are easily damaged by mishandling. Handle the units by their base only.
DO NOT handle units by the top platform or any attachments to the top platform as this may damage the unit.

NOTICE

DO NOT remove transit screws for the PY004VX(/M) platforms. These must remain in place until the actuators are fitted.

2. For the PY004Z9(/M) platforms only, remove the transit inserts which are clearly indicated by yellow stickers on the top surface of the stage. **DO NOT** remove transit screws for the PY004VX(/M) platforms. These must remain in place until the actuators are fitted (see Section 3.3 for setting up).

3.2 Checking the Units

If, when opened, the units are damaged or incorrectly supplied in any way, contact Thorlabs. If a return is required, use the original packaging. If this is not available, use a strong box and surround the stage with shock-absorbent material.

3.3 Setting up the Device

The setup of each PY004 stage is different depending on the device type. Refer to the appropriate section for the device ordered.

NOTICE

The PY004 platforms can be modified with different actuator fittings. To remove existing actuators, see Section 3.3.6.

3.3.1 Setting up the PY004(/M)

There are no specific instructions for setting up the PY004(/M).

The stage is ready for operation straight out of the box.

3.3.2 Setting up the PY004Z9(/M)

There are no specific instructions for setting up the PY004Z9(/M).

The stage is ready for operation straight out of the box.

3.3.3 Setting up the PY004VX(/M)

NOTICE

This procedure applies to all actuator types fitted to the PY004VX(/M) platforms.

Fit the actuators as follows:

1. Loosen the top two (M3 x 0.5) screws with a a BD-2.5M 2.5 mm hex key, on the actuator clamping bracket.
2. Push each actuator into each of the clamping bracket holes, making sure the front flange is correctly seated against the front face on the bracket.
3. Holding the actuator in place, tighten both clamping bracket screws to 22-25 cN·m of torque.
4. Make sure both actuators are securely fitted before proceeding.
5. Do one of the following:
 - a. For manual and PIA actuators, screw in each of the actuators by hand using the thumb nut adjuster until resistance is felt and the screw starts to tighten.

For PIA13 actuators, orientate as shown in Figure 6.

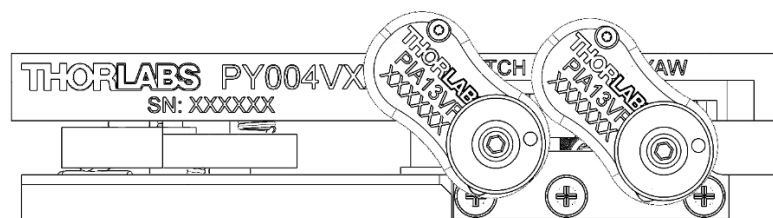


Figure 6 PIA13(VF) Actuator Orientation

NOTICE

The PIA13(/VF) actuators are not compatible with the PY004(/M) and PY004Z9(/M).

- b. For Z9 actuators, set up the Z9 actuator in Kinesis as a linear drive (use the Z9 user manual for reference) and then using Kinesis, drive each actuator to a position of 8.5 mm, then disable the actuator.
6. Turn the unit over, with the top face down.
7. Remove the two brass M3 x 0.5 locking screws using a PH2 driver.
8. Remove the transit label.

NOTICE

Retain the locking screws and transit label for future shipping or changing of actuators.

9. For Z9 actuators only, follow the instructions in Chapter 4 for full stage setup with Kinesis.

The stage is now ready for operation.

3.3.4 Mounting to the Work Surface

The base of the pitch and yaw platform is provided with a series of through holes for attachment to metric or imperial optical tables and breadboards. These holes allow the unit to be mounted with the top platform mounting holes in line with the work surface hole pattern or midway off the hole pattern center (see Figure 7 to Figure 9).

The unit can also be secured off the work surface hole pattern by using two CL series clamps.

When mounting the platform close to other equipment, ensure that the travel of the moving platform is not obstructed.

NOTICE**Damage to equipment**

If the moving platform is driven against a solid object, damage to the internal mechanism could occur.

To check the range of travel on each axis, see Section 1.4.

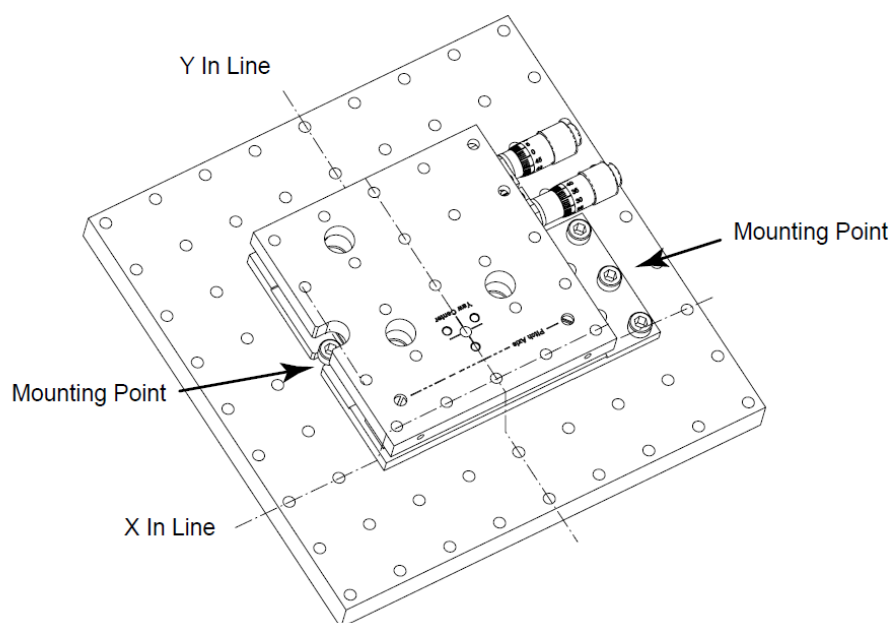


Figure 7 **Mounting X and Y Axes In-Line with Hole Pattern**

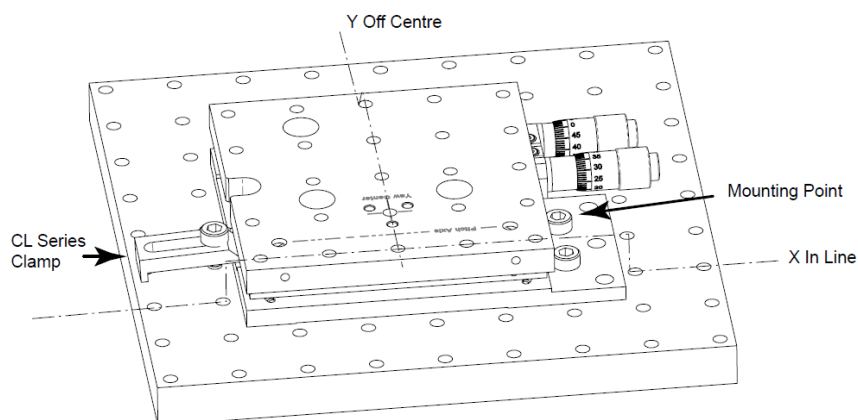


Figure 8 *Mounting X-Axis In-Line with Hole Pattern*

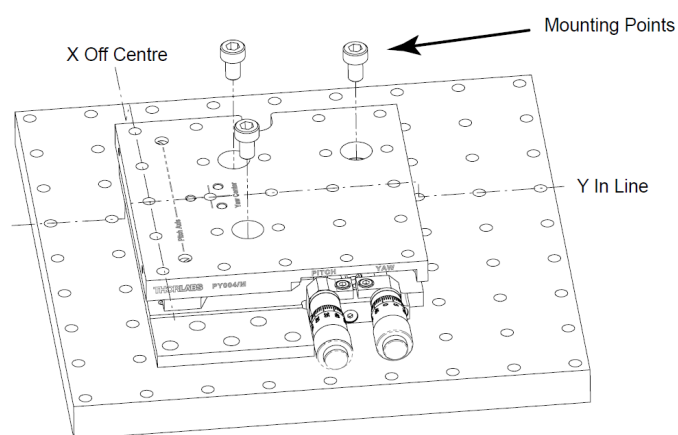


Figure 9 *Mounting Y Axis In-Line with Hole Pattern*

3.3.5 Mounting Equipment to the Moving Platform

The top platform features an array of 24 1/4"-20 (M6 x 1.0) threaded holes on a standard 1" (25 mm) pitch.

Damage to equipment

To avoid damaging the internal mechanism:

NOTICE

- The weight attached to the moving platform must not exceed 5 kg (11 lbs) for manual stages and 2 kg (4.4 lbs) for motorized stages.
- Do not use bolts that extend beyond the bottom surface of the top plate.
- Do not apply excessive force to the moving platform.

3.3.6 Removing Actuators

Existing actuators can be removed and different ones installed.

To remove existing actuators, do the following:

1. Loosen both clamping bracket screws with a BD-2.5M 2.5 mm hex key.
2. Carefully pull each actuator from clamping bracket holes and retain.

Refer to Sections 3.3.1 to 3.3.3 for fitting different actuators.

Chapter 4 Operation with Kinesis Software

NOTICE

This section applies only to the PY004Z9(/M) stages or stages fitted with Z912B(/V) actuators when driven by the latest Kinesis software version, which is available for download on the Thorlabs Website.

4.1.1 Compatible Drivers and Software

The Thorlabs compatible controller for motorized PY004 stages fitted with Z912B(/V) actuators is the K-Cube Single Channel DC Motor Controller KDC101.

4.1.2 Setting Up the System

Several parameters must first be set to ensure that a particular stage is driven properly by the system. These parameters relate to the physical characteristics of the stage being driven (for example, min and max positions, leadscrew pitch, homing direction, and so forth). To assist in setting these parameters correctly, a pop-up window appears when the Kinesis software is first started, prompting the user to select a specific stage type and axis driven by the motor controller. Once this association has been made, the Kinesis software automatically applies suitable default parameter values upon software launch.

NOTICE

If the stage type is not associated on startup, the software will associate a Z906 type actuator by default.

Do the following:

1. Install the electronic hardware and connect the controllers to the relevant axes of the associated stage/actuators (refer to the manual supplied with the KDC Controller).

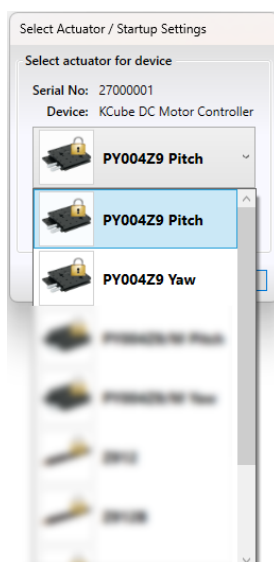
NOTICE

Two controllers are needed; one for each axis.

2. Shut down all applications using the Kinesis server.
3. Open Kinesis.

On startup, the **Select Actuator / Startup Settings** window is displayed to select the stage driven by the Z912 actuator.

4. In the **Select Actuator / Startup Settings** window, select the small black arrow to display the drop-down box and select **PY004Z9 Pitch** or **PY004Z9 Yaw** as appropriate and click OK.



A default configuration is set at the factory and stored in the motor controller's non-volatile memory. When the stage type is selected, the server reads this stage and controller information. The Home position for each axis is set in the middle of the range of motion.

Refer to the manual supplied with the DC motor controller for further information.

4.1.3 Homing the Motors

Homing the motor moves the actuator to the home limit switch and then to its central position (as dictated by the zero offset parameter in the settings panel). The internal position counter and the GUI display are then set to zero. The limit switch provides a fixed datum that can be found after the system has been powered up.

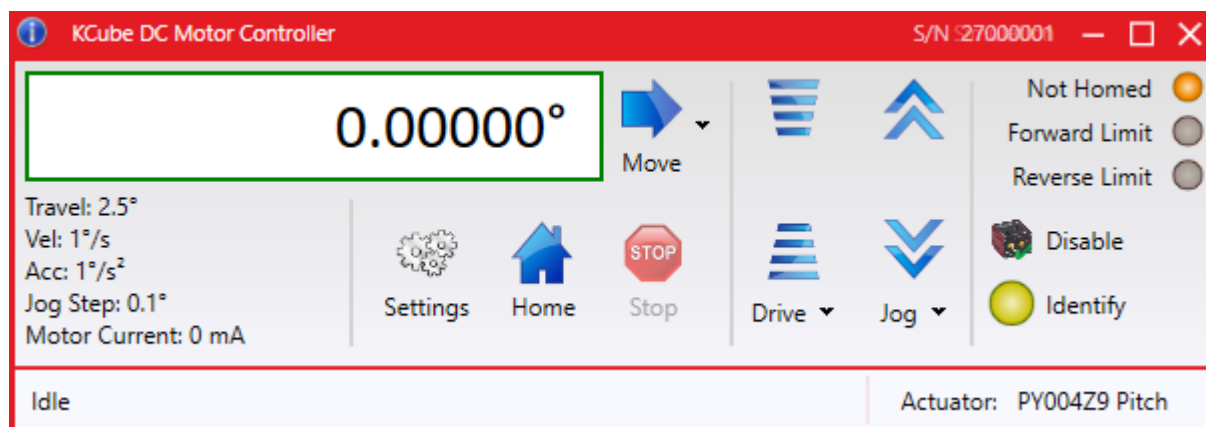


Figure 10 K Cube DC Motor Controller Kinesis Software GUI

To home the motors from the Kinesis software GUI, select **Home**.

The LED labeled **Not Homed** changes to **Homing** and when homing is complete, changes again to **Homed** and the LED indicator becomes green. The displayed position counts down to 0° (the home position).

NOTICE

Homing can also be performed from the top panel settings menu. Refer to the KDC101 driver manual for more details.

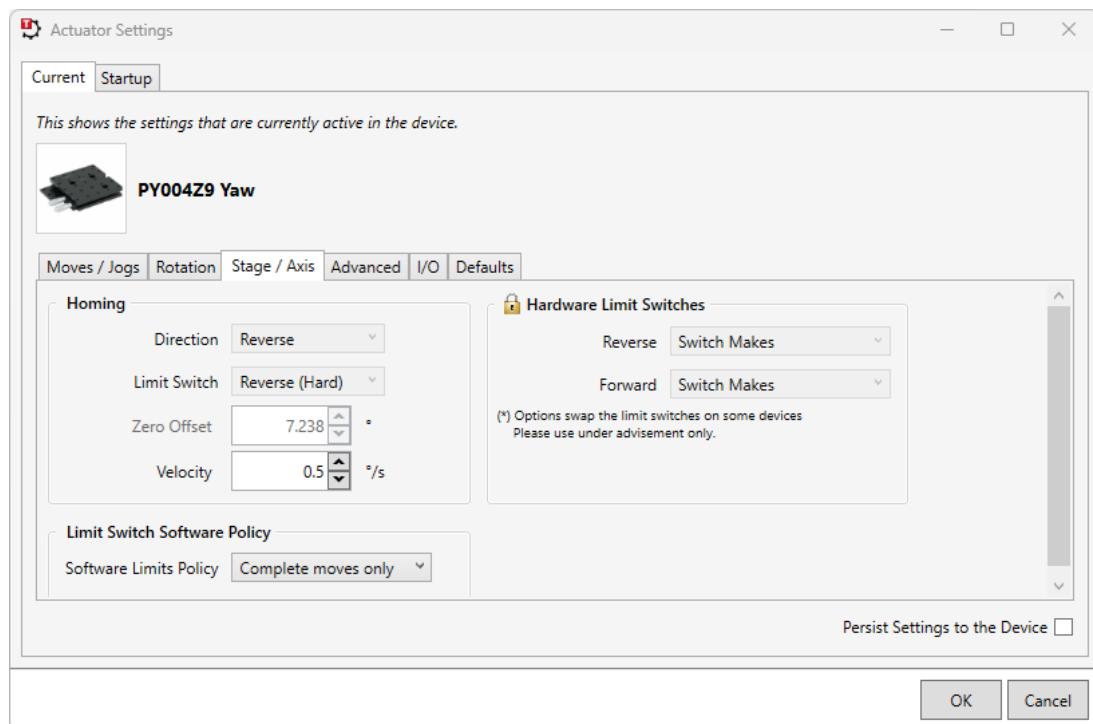
NOTICE

Due to manufacturing tolerances, a small position offset (misalignment of the top and bottom platforms) may be visible after homing. This offset can be removed in the Settings panel by creating a set of custom device startup settings. Refer to Section 4.1.4.

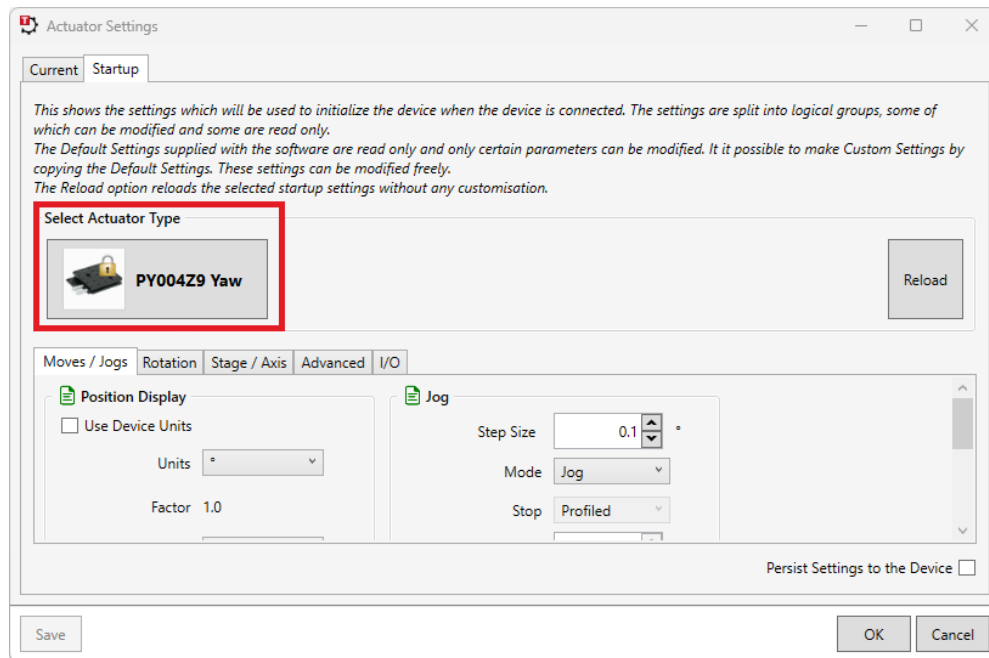
4.1.4 Creating Custom Startup Settings

After homing, a small position offset (misalignment of the top and bottom platforms) may be visible. To remove this, do the following:

1. Select the **Jog** buttons to align the stage as required.
2. Note the value shown in the GUI display.
3. From the **KCube DC Motor Controller** window select the Settings icon  to display the **Actuator Settings** window. From the **Current** window select the **Stage / Axis** tab. The misalignment value is displayed in the **Zero Offset** field.

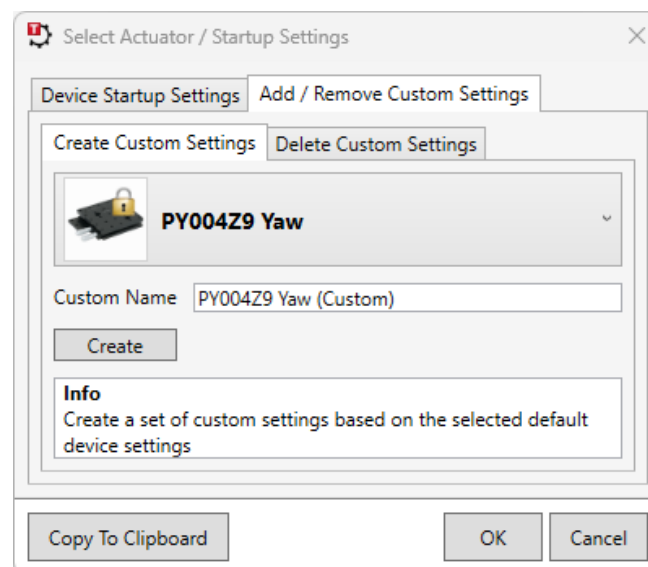


4. From the **Actuator Settings** window, select the **Startup** tab and select the device icon displayed immediately below the **Select Actuator Type** heading.



The **Select Actuator / Startup Settings** window showing the default settings for the selected actuator is displayed.

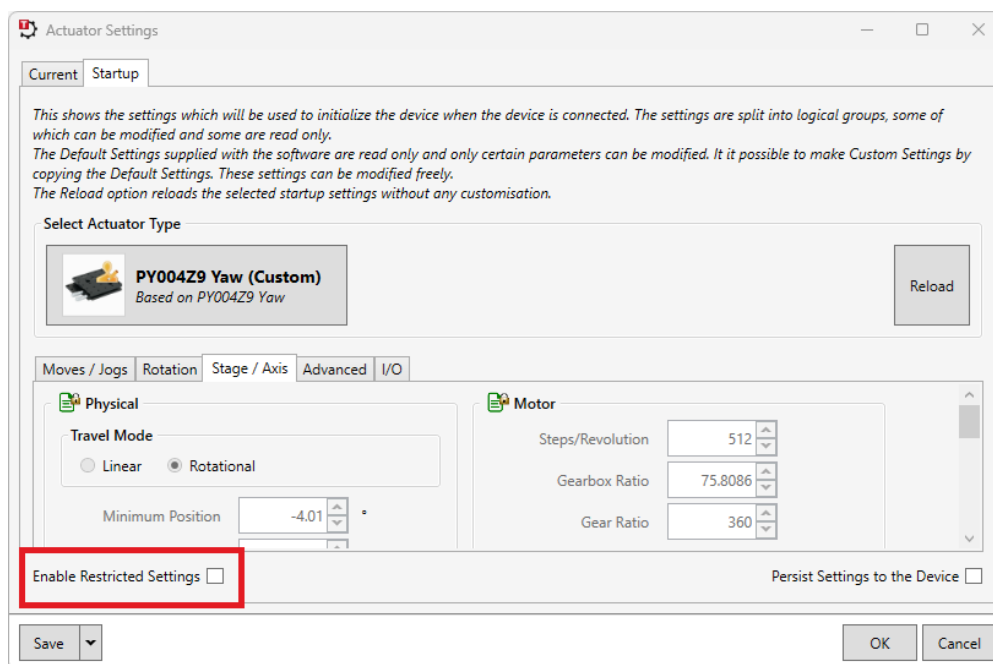
5. Select the **Add / Remove Custom Settings** tab.



6. In the **Custom Name** field, enter a suitable name (for example, "PY004Z9 Yaw (Custom)") and select **Create**.
7. Select **OK**.

The custom settings are saved.

8. From the **Actuator Settings** window, select the device icon displayed immediately below the **Select Actuator Type** heading.
9. From the drop-down menu, select the custom setting created in the previous steps and select **OK**.
10. Select the **Stage/Axis** tab and check the **Enable Restricted Settings** checkbox to enable editing of the restricted parameters. This is a safeguard against accidentally editing restricted parameters in the Custom Device Startup Settings mode.



A confirmation message window is displayed.

11. In the confirmation message window, select **Yes**.
12. Add the value noted in step 2 to the value currently shown in the **Zero Offset** field and enter this new value.
13. Select **Save**, then **OK** to save the new setting.
14. Repeat steps 1 to 13 for the other axis of the stage.

Subsequent home moves will move to the new aligned positions.

Chapter 5 Maintenance

The following preventative maintenance procedures should be done periodically:

- Regreasing the lead screws.
- Replacing the actuators.

5.1 Regreasing the Lead Screws

NOTICE

Regrease only standard Z9 motors. This procedure is NOT suitable for PIA actuators, manual screws, or vacuum Z9 motors.

Regreasing should be performed whenever the lead screw thread appears dry, or the motor is noisy. Periodic greasing is advised in applications with a high-duty cycle, typically after 4000 cycles in pitch and/or 1800 cycles in yaw.

Do the following:

1. Drive both motors to extend the lead screws to their full extent.
2. Apply a thin bead of high load, silicon free lubricant (for example, GKZ8 grease).

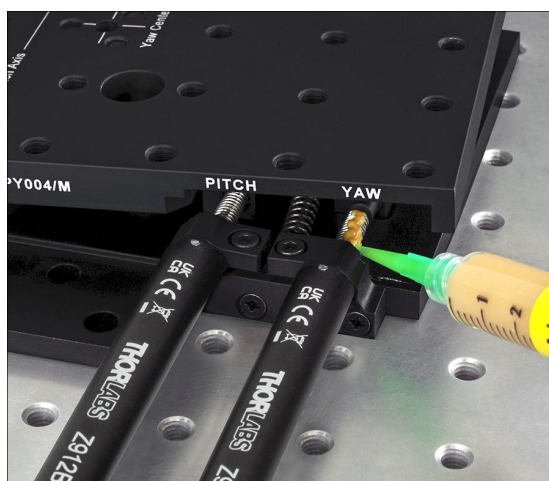


Figure 11 Greasing the Leadscrew

3. Drive the motor to retract the lead screw and distribute the grease.
4. Remove excess grease using a lint free cloth.

5.2 Replacing the Actuators

When the actuators show signs of wear, they should be replaced.

When replacing/changing the actuators, the tightening torque of the barrel clamps must be more than 22 cN·m and must not exceed 25 cN·m.

Refer to Section 3.3.6 for the procedure to remove actuators.

Chapter 6 Thorlabs Worldwide Contacts

For technical support or sales inquiries, please visit us at www.thorlabs.com/contact for our most up-to-date contact information.



Corporate Headquarters

Thorlabs, Inc.
43 Sparta Ave
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